Introduction to Wireless Sensor Networks

Outline

- **Introduction**
- **Applications**
- **Characteristics**
- **Challenges**

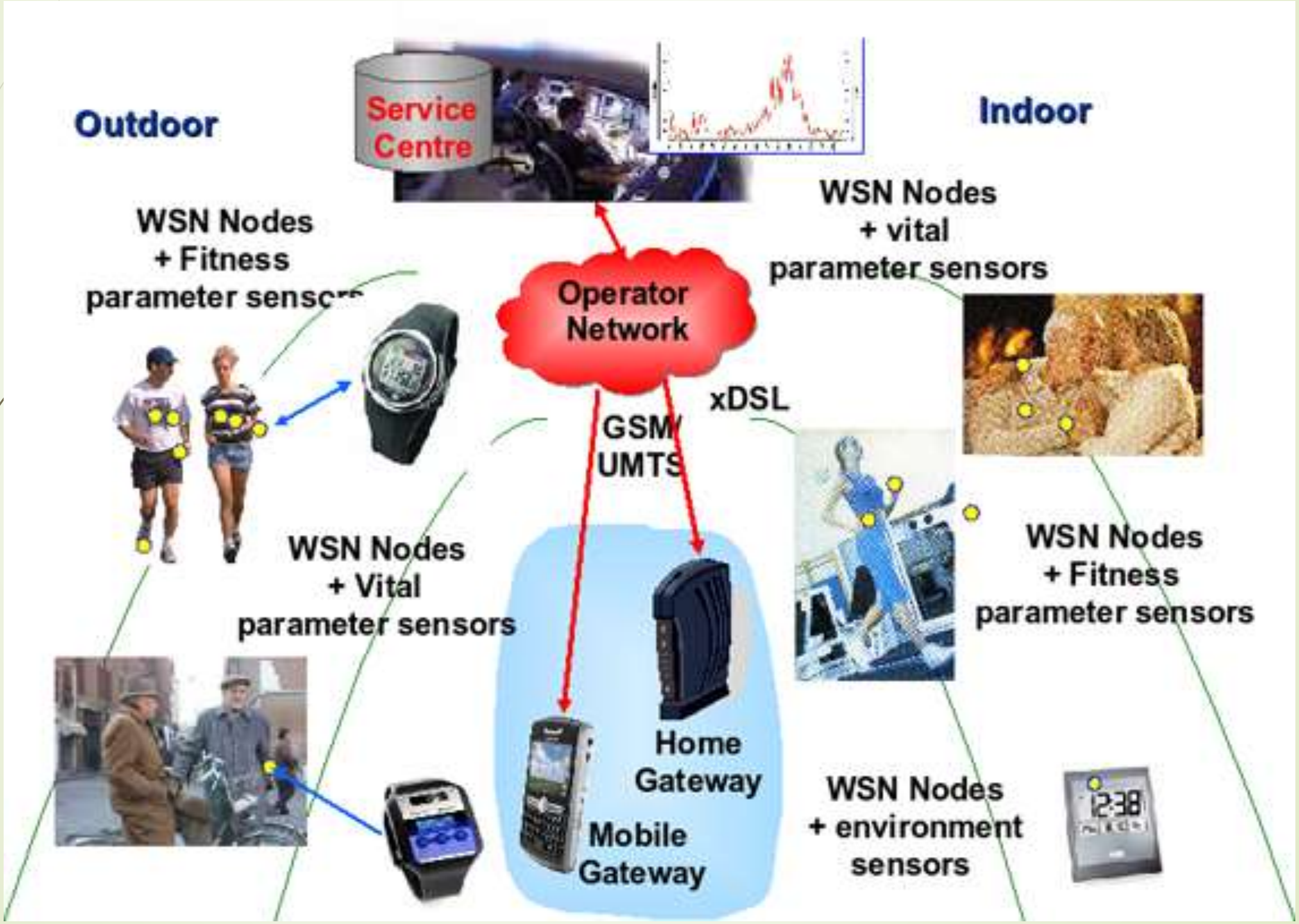
Introduction

- Wireless Sensor Networks are networks that consists of sensors which are distributed in an ad hoc manner.

- These sensors work with each other to sense some physical phenomenon and then the information gathered is processed to get relevant results.

- Wireless sensor networks consists of protocols and algorithms with self-organizing capabilities.

Example of WSN



Applications of Wireless Sensor networks

The applications can be divided in three categories:

1. Monitoring of objects.
2. Monitoring of an area.
3. Monitoring of both area and objects.

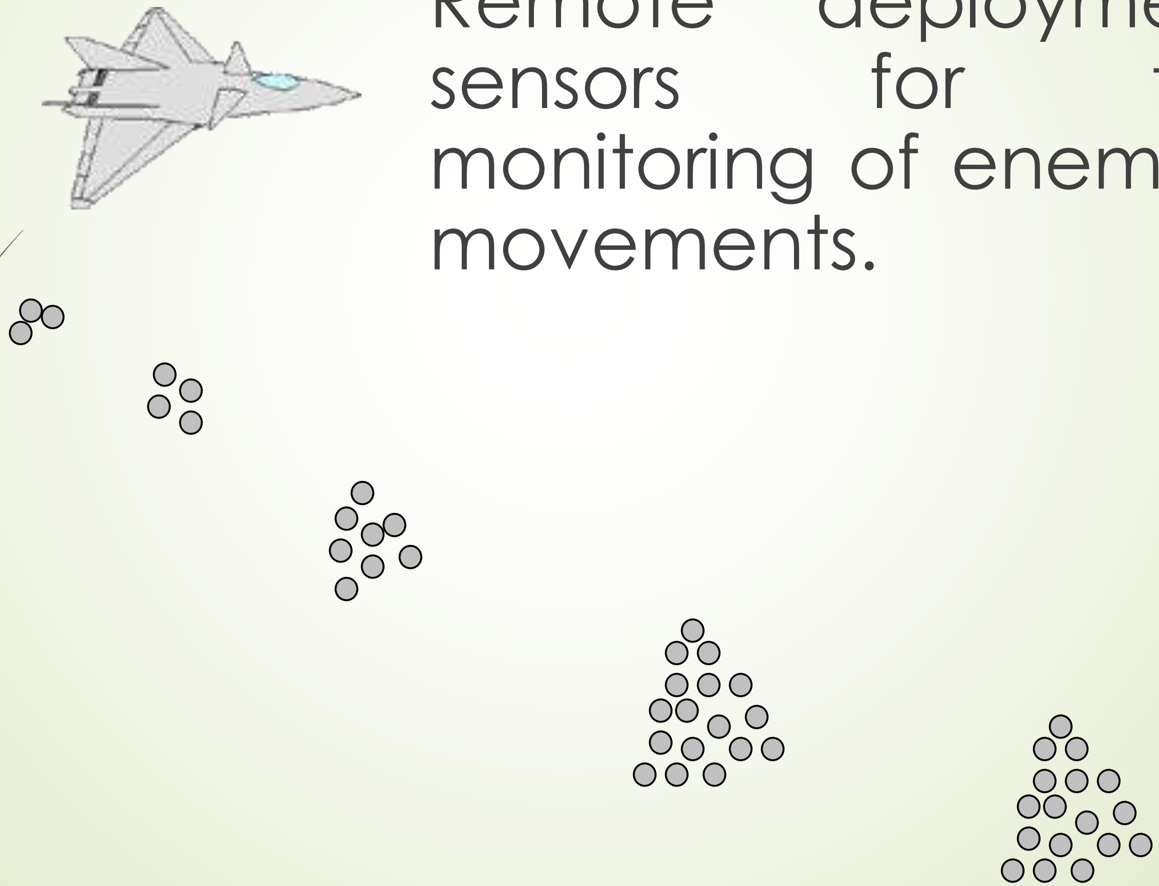
Monitoring Area

- Environmental and Habitat Monitoring
- Precision Agriculture
- Indoor Climate Control
- Military Surveillance
- Treaty Verification
- Intelligent Alarms

Monitoring Objects

- Structural Monitoring
- Eco-physiology
- Condition-based Maintenance
- Medical Diagnostics
- Urban terrain mapping

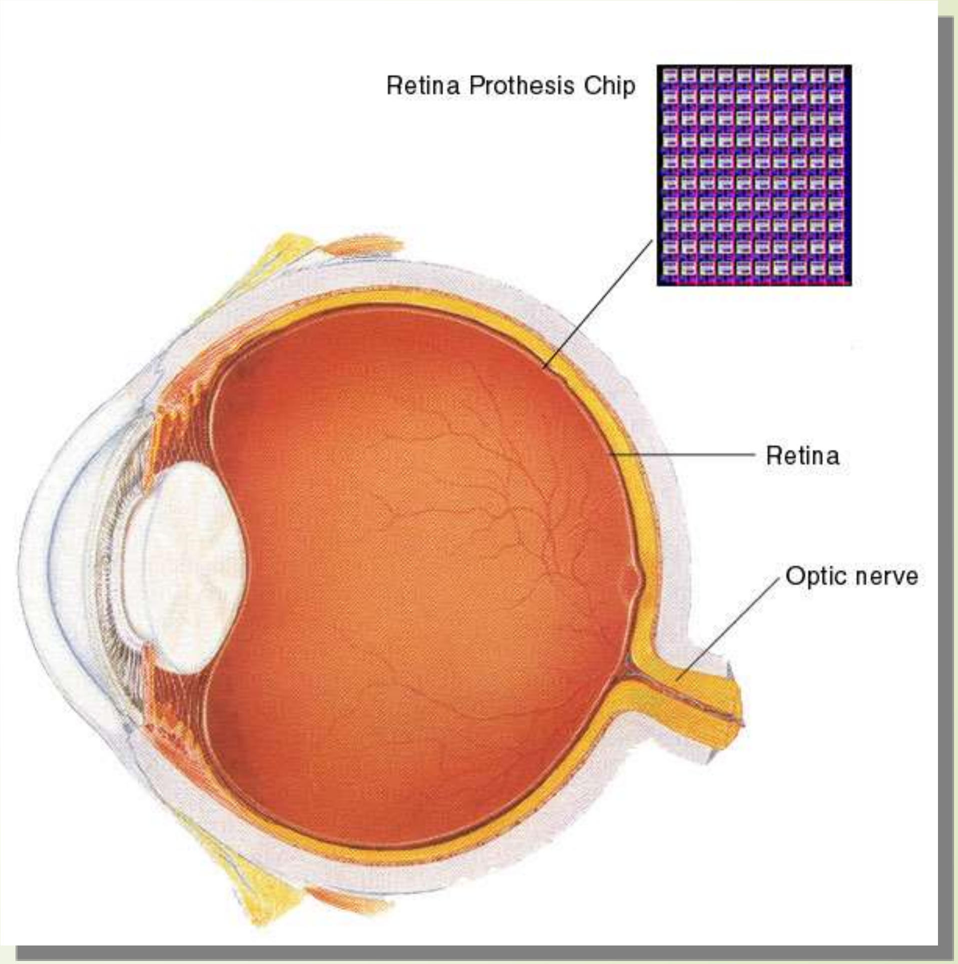
Military



Remote deployment of sensors for tactical monitoring of enemy troop movements.

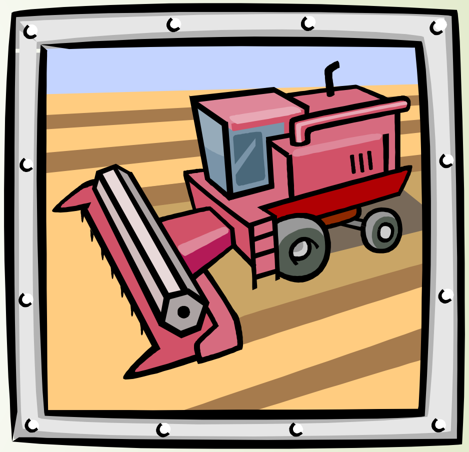
Biomedical / Medical

- Health Monitors
  - Glucose
  - Heart rate
  - Cancer detection
- Chronic Diseases
  - Artificial retina
  - Cochlear implants
- Hospital Sensors
  - Monitor vital signs
  - Record anomalies



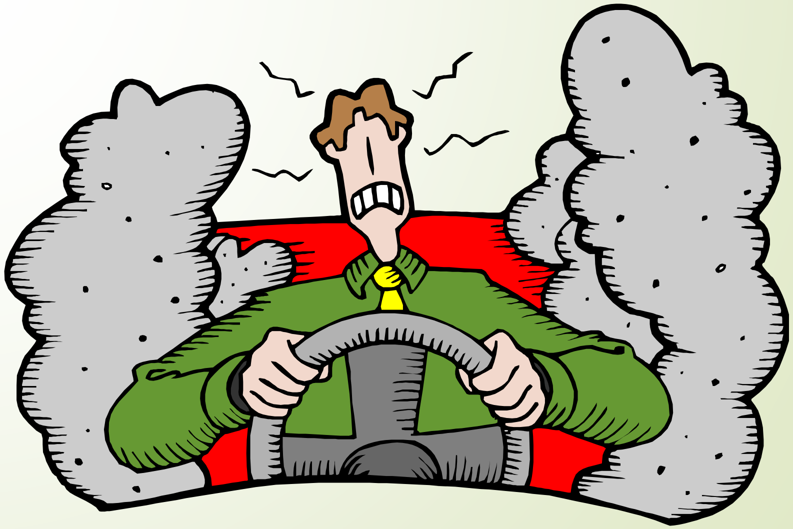
Industrial & Commercial

- Numerous industrial and commercial applications:
  - Agricultural Crop Conditions
  - Inventory Tracking
  - In-Process Parts Tracking
  - Automated Problem Reporting
  - RFID – Theft Deterrent and Customer Tracing
  - Plant Equipment Maintenance Monitoring



Traffic Management & Monitoring

- Future cars could use wireless sensors to:
  - Handle Accidents
  - Handle Thefts



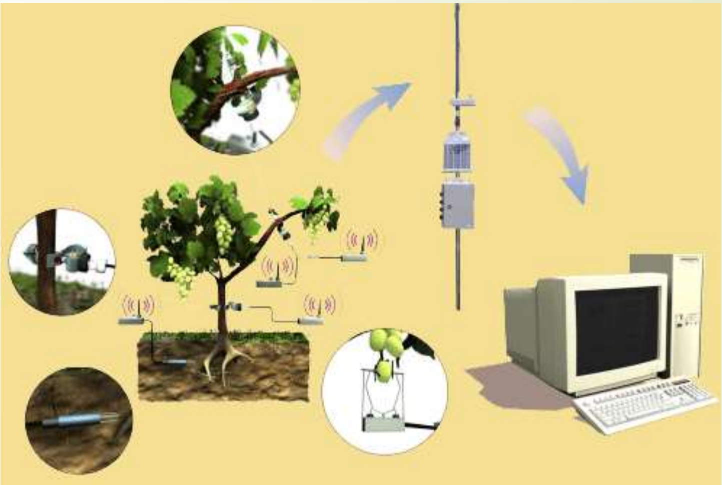
✓Sensors embedded in the roads to:

- –Monitor traffic flows
- –Provide real-time route updates

Example: Precision Agriculture

- Precision agriculture aims at making cultural operations more efficient, while reducing environmental impact.

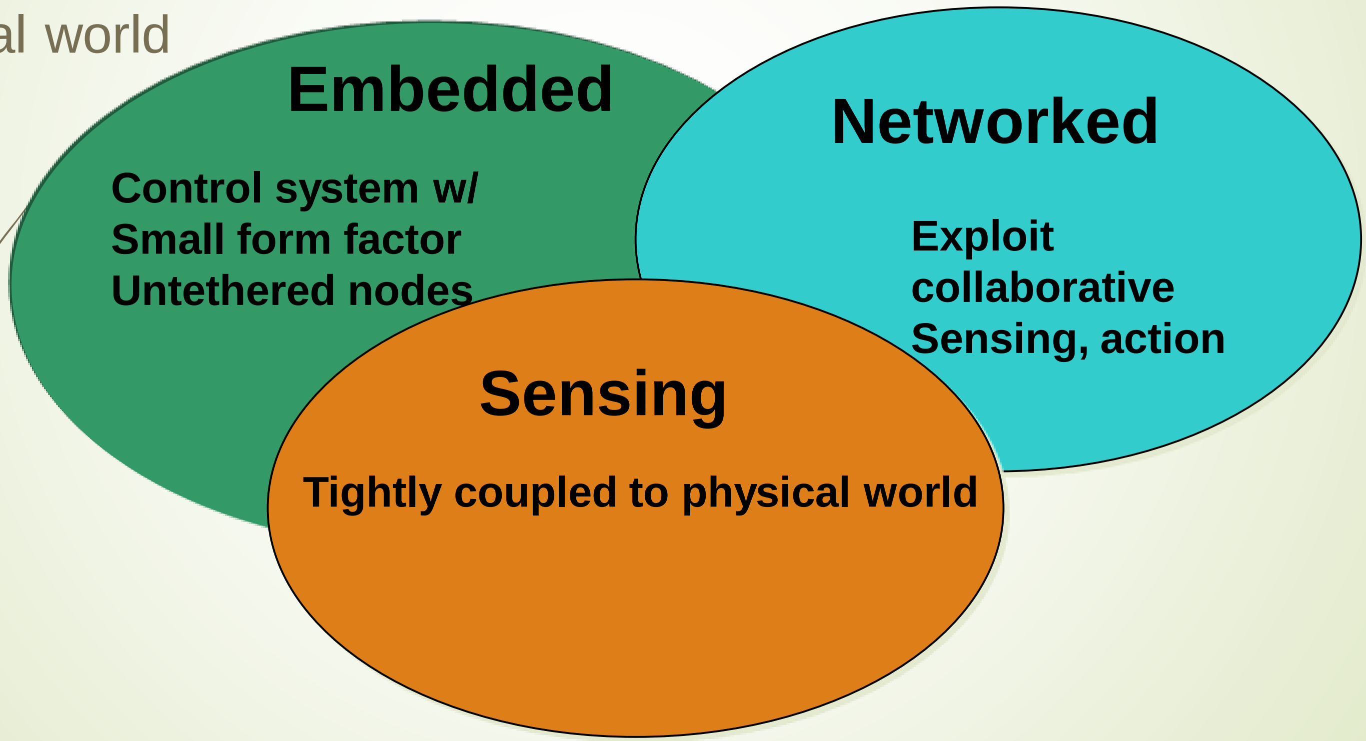
- The information collected from sensors is used to evaluate optimum sowing density, estimate fertilizers and other inputs needs, and to more accurately predict crop yields.



Enabling Technologies

Embed numerous distributed devices to monitor and interact with physical world

Network devices to coordinate and perform higher-level tasks



Exploit spatially and temporally dense, in situ, sensing and actuation

Design Challenges

- **Heterogeneity**
  - The devices deployed maybe of various types and need to collaborate with each other.
- **Distributed Processing**
  - The algorithms need to be centralized as the processing is carried out on different nodes.
- **Low Bandwidth Communication**
  - The data should be transferred efficiently between sensors

Characteristics of Wireless Sensor Networks

- Wireless Sensor Networks mainly consists of **sensors. Sensors** are -
  - low power
  - limited memory
  - energy constrained due to their small size.
- Wireless networks can also be deployed in **extreme environmental** conditions and may be prone to enemy attacks.
- Although deployed in an ad hoc manner they need to be **self organized** and **self healing** and can face constant reconfiguration.

Continued..

- **Large Scale Coordination**
  - The sensors need to coordinate with each other to produce required results.
- **Utilization of Sensors**
  - The sensors should be utilized in a ways that produce the maximum performance and use less energy.
- **Real Time Computation**
  - The computation should be done quickly as new data is always being generated.

Operational Challenges of Wireless Sensor Networks

- Energy Efficiency
- Limited storage and computation
- Low bandwidth and high error rates
- Errors are common
  - Wireless communication
  - Noisy measurements
  - Node failure are expected
- Scalability to a large number of sensor nodes
- Survivability in harsh environments
- Experiments are time- and space-intensive

Comparison with ad hoc networks

- Wireless sensor networks mainly use **broadcast** communication while ad hoc networks use **point-to-point** communication.
- Unlike ad hoc networks wireless sensor networks are **limited by sensors** limited power, energy and computational capability.
- Sensor nodes may **not have global ID** because of the large amount of overhead and large number of sensors.

References

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